

Why the 27×27 cubic system has an analytic inverse:
the commutant of O_h on $\mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \mathbb{R}^3$

Abstract

The dynamic single-site system for a cubic voxel is a 27×27 matrix M carrying 19 independent entries. Inverting it by brute force produces an expression of no use to anybody. This note explains why that is the wrong thing to do: M commutes with the octahedral group acting on $\mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \mathbb{R}^3$, and Schur's lemma then forces M into block-diagonal form in a basis fixed by the *symmetry alone*. The parameters $a, \dots, t$ move the eigenvalues; they cannot move the eigenvectors. Inversion therefore reduces to inverting a handful of small blocks, analytically and in general moduli.

1 The space and the group

The unknowns are *raw* derivatives — $\partial_p \partial_q u_i$ with all three indices free and no symmetrisation — so the solution space is

$$V = \mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \mathbb{R}^3, \quad \dim V = 27.$$

This is worth stating plainly because it is *not* the space used by the Voigt-based T_{27} , which is $3 \oplus 6 \oplus 18$ (displacement, symmetric strain, quadratic) and projects the antisymmetric part out. The two spaces have the same dimension and different content.

A cube is invariant under the octahedral group O_h , realised concretely as the 48 signed permutation matrices — every 3×3 matrix with exactly one ± 1 in each row and column. Because V is a threefold tensor power, each $R \in O_h$ acts on it as

$$\rho(R) = R \otimes R \otimes R.$$

Verified. M commutes with ρ on the generators of O_h — a 90° rotation about z , a 120° rotation about (111) , and the mirror $z \mapsto -z$:

$$\rho(R) M \rho(R)^{-1} - M = 0 \quad \text{identically in } a, \dots, t.$$

Checked symbolically, not numerically: the difference is the zero matrix as a polynomial identity in the nineteen parameters, so it holds for every medium and every frequency, not at a sampled point.

2 The consequence

Theorem 1 (Schur). *Let $V = \bigoplus_\lambda \Gamma_\lambda \otimes \mathbb{C}^{m_\lambda}$ be the isotypic decomposition of V under G , where Γ_λ is an irreducible representation of dimension d_λ occurring with multiplicity m_λ . Then every G -equivariant operator M has the form*

$$M = \bigoplus_\lambda \text{id}_{\Gamma_\lambda} \otimes B_\lambda, \quad B_\lambda \in \mathbb{C}^{m_\lambda \times m_\lambda},$$

and the commutant algebra is $\text{End}_G(V) \cong \bigoplus_{\lambda} \mathbb{C}^{m_{\lambda} \times m_{\lambda}}$, of dimension $\sum_{\lambda} m_{\lambda}^2$.

The content for us is the placement of the tensor factors. M acts as the *identity* on the internal index of each irreducible representation and as an $m_{\lambda} \times m_{\lambda}$ matrix on the multiplicity space. Two things follow, and they are what make the problem tractable.

Corollary 2 (The basis is parameter-free). *A symmetry-adapted basis block-diagonalises M for all values of $a, \dots, t$ simultaneously. The nineteen parameters enter only through the entries of the small matrices B_{λ} ; they cannot rotate the blocks, because the blocks are determined by O_h before M is known.*

Corollary 3 (Inversion is small).

$$M^{-1} = \bigoplus_{\lambda} \text{id}_{\Gamma_{\lambda}} \otimes B_{\lambda}^{-1},$$

so inverting a 27×27 matrix costs exactly the inversion of the B_{λ} , whose sizes are the multiplicities m_{λ} , each result reused d_{λ} times. A brute-force symbolic inverse of M recomputes the same small quantities 27 times over, in a basis that hides them.

3 What is measured

quantity	value
$ O_h $	48
$\dim \text{End}_G(V) = \sum_{\lambda} m_{\lambda}^2 = \frac{1}{48} \sum_R (\text{tr } R)^6$	31
free parameters in M	19
generic eigenvalue degeneracies of M	$\{1, 2, 3^{\times 8}\}$

The commutant dimension is computed exactly from the character $\chi(R) = (\text{tr } R)^3$ of ρ , with no character table required, since $\sum_{\lambda} m_{\lambda}^2 = \langle \chi, \chi \rangle$.

Two readings of this table matter.

M does not fill its commutant. Nineteen parameters inside a 31-dimensional algebra means M carries 12 further relations beyond O_h -equivariance. Those are physics, not symmetry — reciprocity and the structure of the Green operator — and they are worth identifying, because each one further constrains the B_{λ} .

4 The decomposition

Everything appearing is *ungerade*, and this needs no computation: $\rho(-I) = (-I) \otimes (-I) \otimes (-I) = -\mathbb{1}_{27}$, so inversion acts as -1 on all of V and every gerade irreducible is excluded. Computed multiplicities confirm it — A_{1g} , T_{1g} , T_{2g} and the symmetric-traceless $E_g \oplus T_{2g}$ all return 0.

The characters were built explicitly rather than copied from a table, each verified irreducible by $\langle \chi, \chi \rangle = 1$: $A_{1u} = \det R$; $A_{2u} = \text{sgn}(P) \det R$ with P the permutation part; $T_{1u} = \text{tr } R$ (the defining representation); $T_{2u} = T_{2g} \cdot \det R$ with $T_{2g} = \frac{1}{2}((\text{tr } R)^2 - \text{tr } R^2) \text{sgn}(P)$; and $E_u = E_g \cdot \det R$ with E_g the symmetric-traceless part less T_{2g} . Then

$$T_{1u} \otimes T_{1u} \otimes T_{1u} = A_{1u} \oplus A_{2u} \oplus 2E_u \oplus 4T_{1u} \oplus 3T_{2u}$$

irreducible	d_λ	m_λ	block B_λ	copies
A_{1u}	1	1	1×1	1
A_{2u}	1	1	1×1	1
E_u	2	2	2×2	2
T_{2u}	3	3	3×3	3
T_{1u}	3	4	4×4	3
$\sum m_\lambda d_\lambda = 27$			$\sum m_\lambda^2 = 31$	

Both totals are independent checks and both are met: 27 is $\dim V$, and 31 was obtained twice by separate routes — the character inner product $\frac{1}{48} \sum_R (\text{tr } R)^6 = 1488/48$, and an exact linear solve for the commutant. The count of five irreducibles matches the dimension of the centre of the commutant, computed independently.

Consistency with the eigenvalue probe. The degeneracies $\{1, 2, 3^{\times 8}\}$ follow: one eigenvalue of degeneracy 1 from each of A_{1u} and A_{2u} , two of degeneracy 2 from E_u , and $4 + 3 = 7$ of degeneracy 3 from T_{1u} and T_{2u} — eleven distinct eigenvalues in all. The probe reported ten because one degeneracy-1 and one degeneracy-2 eigenvalue fell within its clustering tolerance and were counted as a single degeneracy-3 value. The exact characteristic polynomial shows the same near coincidence. This is why the decomposition is settled by characters and by the commutant, not by a degeneracy count.

The practical result.

$$M^{-1} = \bigoplus_{\lambda} \text{id}_{\Gamma_\lambda} \otimes B_\lambda^{-1}, \quad \text{blocks } 1 \times 1, 1 \times 1, 2 \times 2, 3 \times 3, 4 \times 4.$$

The largest object to invert symbolically is a 4×4 . That is closed form at general λ, μ and general frequency, and it is the whole content of the 27×27 inverse.

5 Why this is the right route

The practical claim is not that a symbolic inverse is unobtainable otherwise — one exists, at roughly a megabyte of expression. It is that such an object answers no question. The block form answers several:

- each B_λ^{-1} is small enough to read, so the frequency and modulus dependence of each symmetry channel is visible;
- a channel that misbehaves (the shear channel, in this project's history) is isolated in one block rather than smeared over 729 entries;
- the static limit can be taken blockwise and checked against the independently derived static system;
- the construction is valid at general λ, μ , so it is not tied to the Poisson-solid values at which the existing inverse was demonstrated.